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BSCS - 3B

Chosen ML Application - Social Media Recommendation

Part A - Identify the ML Problem

1. **What is the application?**
   The application is a Social Media Recommendation System. This is the type of system used by social media platforms to suggest videos, posts, or other content based on what a user usually watches or interacts with.

2. **What problem are you trying to solve?**
   The main problem is how to recommend content that a user will probably enjoy or find interesting. Since there are a lot of posts uploaded every day, it would be difficult for users to find everything they like by themselves. A recommendation system can help by showing content that matches their interest.

3. **What are the inputs?**
   Some of the inputs are the user's likes, comments, searches, videos they watch, how long they watch the video, and the accounts or pages they follow. These activities can give information about the user's interests and the type of content they usually interact with.

4. **What is the expected output?**
   The expected output is a set of recommended posts or videos. For example, if a user often watches dancing videos, the system may recommend more dancing related videos. The goal is to give the user content that they are more likely to watch or interact with.

5. **What patterns or relationships might the ML system learn?**
   The system may learn that certain user actions are connected to certain interests. For example, if a user regularly watches and likes volleyball videos, the system may learn that sports related content is interesting to that user. It can also notice which types of posts the user usually skips or watches for a longer time.

6. **What data would the system need?**
   The system would need data such as the user's viewing history, likes, comments, searches, follows, and interactions with different posts. It could also use information about the content itself, such as its topic or category. More examples of user activity can help the system recognize patterns more accurately.

Part B - Traditional Programming vs. Machine Learning

Traditional Programming
In Traditional Programming, the developer creates the rules or guidelines that the system needs to follow. For example, If a user likes many gaming videos, the program or the system could be instructed to show more gaming videos. The developer would have to manually create many rules for different types of users and content.

Input → Algorithm or Rules → Output

Machine Learning
In Machine Learning, the system is not given every rule manually. Instead it is given data from previous user activities and learns patterns from those examples. For example, It can look at what a user watches, likes, and searches for, then learns what type of content that user is likely to interact with. The system or the program can use the pattern it discovered to recommend new content after learning from the data.

Input or Data → MLalgorithm → Learned Pattern → Prediction

The main difference is that Traditional Programming depends on rules that are created by the developer. On the other hand, Machine Learning uses data to learn patterns and make predictions.

Part C - Create Your Own Mini Dataset

Time Spent Watching Videos and Recommendation Score

Time Spent Watching (minutes)	Recommendation Score
5	1
10	2
15	3
20	4
25	5

1. **Identify the Patterns**
   The pattern is that the output increases by 1 whenever the input increases by 5.
2. **Determine the missing output**
   If the input is 30 minutes, the output is 6.
3. **Write the mathematical relationship/function.**

The relationship is Output = Input / 5
Or $y = x / 5$

4. **Explain how a machine learning algorithm could learn this relationship**
   The algorithm can compare the given input and output examples and observe how they change. From the examples, it can learn that the output increases by 1 for every additional 5 minutes. After learning this pattern, it can use the relationship to predict the output for a new input.

Part D - Explain the Training Process

The machine learning algorithm is given data with examples of inputs and outputs during training. It studies these examples and looks for patterns between them. For example, the program can learn that when a user spends more time watching a certain type of content, their interest in that content may also increase. After learning from enough examples, the algorithm uses the patterns it learned to make predictions for new data.

Data → Training → Pattern Learning → Prediction

Short Reflection

I learned that machine learning is different from normal programming because it does not always need a developer or a programmer to create a rule for every situation. Instead, it can learn from examples and find patterns in the data. I can relate to this because whenever I search for something on TikTok or Facebook, sometimes I notice that shops, products, or recommendations related to what I searched for start appearing on my feed. This shows how the system can use my previous searches and activities to recommend things that I might be interested in.

I also learned that machine learning can be useful because it can make our tasks easier and more personalized. Instead of manually choosing what content to show to every user, the system can learn from their activities and make recommendations based on their interest. However, the recommendations still depend on the data that the system receives, so having enough and useful data is important.